EE 4091 - Photonics

Assignment 3

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Part I :

(2) Light output power → 2.5mW
Current → 50mA
Voltage → 1.6V

$P = I \times V = 50mA \times 1.6V$

$P = 0.08\,W$

$$\text{LED external efficiency} = \frac{\text{Light output power}}{\text{Electrical power output}} \times 100 = \frac{0.0025\ W}{0.08\ W} \times 100 = 3.125\%$$

Part II :

(1)

Cavity length (L) = 1m
Mirror refractivities (R) = 0.99
Wavelength of operation (λ) = 632.8nm
refractive index (n) = 1

- mode number (m) = ?
- free spectral width (ΔV) = ?
- Quality factor (Q) = ?
- Finesse (F) = ?

• mode number:

$$m = \frac{L}{(\lambda/2)} = \frac{1}{(632.8\times10^{-9}/2)} = 3.16\times10^{6}$$

• free spectral width:

$$\Delta V = \frac{c}{2L} = \frac{3\times10^{8}}{2} = 1.5\times10^{8}\ Hz$$

• Finesse: $F = \frac{\pi Q}{2} = \frac{\pi \times 987.81\times10^{6}}{2}$

• Quality factor:

$$Q = \frac{\pi m R}{1-R} = \frac{\pi \times 3.16\times10^{6} \times 0.99}{0.01}$$

$F = 1.54\times10^{9}$

$Q = 982.81\times10^{6}$

(2)

n = 3.45
λ = 1300 nm
Laser gain spectrum width = 8.8nm

Cavity length = ? $L = \frac{\lambda}{2n} = \frac{1300\times10^{-9}}{2\times3.45} = 188.405\,nm$

Comment:

One longitudinal mode is supported within the 8.8nm laser gain spectrum width thanks to the predicted cavity length. This design helps in single-mode operation and prevents spectrum widening and undesirable mode competition.

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Part III

I_p: Photocurrent
R: Responsivity

① Optical power $P = \frac{I_p}{R}$

for InGaAs:

$$P = \frac{10nA}{0.9\ A/W} = \frac{10\times10^{-9}\ A}{0.9\ A/W} = 11.11\times10^{-9}\ W$$

$P = 11.11\ nW$ (InGaAs)

for GaP:

$$P = \frac{10nA}{0.09\ A/W} = 111.11\ nW$$

for Si(C):

$$P = \frac{10nA}{0.58\ A/W} = 17.24\ nW$$

for Si(E):

$$P = \frac{10nA}{0.6\ A/W} = 16.66\ nW$$

for Ge:

$$P = \frac{10nA}{0.95\ A/W} = 10.52\ nW$$

② Quantum efficiency

$$\eta = \frac{\text{ehp flux making the optical current}}{\text{incident photon flux } (\Phi)} = \frac{I/e}{P/hc/\lambda}$$

$$\frac{hc}{\lambda} = 1.28\times10^{-19}$$

$$\eta \text{ for InGaAs} = \frac{10\times10^{-9}/1.6\times10^{-19}}{11.11\times10^{-9}/1.28\times10^{-19}} = 0.72 = 72\%$$

$$\eta \text{ for Ge} = \frac{10\times10^{-9}/1.6\times10^{-19}}{10.52\times10^{-9}/1.28\times10^{-19}} = 0.89 = 89\%$$

$$\eta \text{ for Si(E)} = \frac{10\times10^{-9}/1.6\times10^{-19}}{16.66\times10^{-9}/1.28\times10^{-19}} = 0.48 = 48\%$$

$$\eta \text{ for Si(C)} = \frac{10\times10^{-9}/1.6\times10^{-19}}{17.24\times10^{-9}/1.28\times10^{-19}} = 0.46 = 46\%$$

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